

AI61201: Visual Computing with AI/ML (Autumn 2026)

Programming Assignment 2: Spatial Domain Image Transformations (20 Marks)

Task 1: Directional Unsharp Masking (6 Marks)

Use image_one.jpg ( Assignment-2 resources)

- **1.1 Filter Design & Visualisation (3 Marks):** Design a Gaussian smoothing filter that performs directional smoothing to facilitate vertical edge sharpening using highboost filtering.
 - Construct an anisotropic 2D Gaussian kernel of size 61×61 . The smaller standard deviation is set to $\sigma = 0.5$, and the ratio of the larger standard deviation to the smaller one is 40: 1. Determine whether the kernel needs to be oriented horizontally or vertically to enhance vertical edges.
 - Plot the level sets (contour plot) and display the heatmap of the designed Gaussian kernel.
- **1.2 Spatial Filtering (1 Mark):** Perform 2D convolution on the input image using `cv2.filter2D()` with your custom anisotropic kernel. Display the filtered (blurred) output.
- **1.3 Unsharp Masking & High Boost Filtering (2 Marks):**
 - Compute the unsharp mask: $Mask = InputImage - BlurredImage$.
 - Apply high boost filtering: $SharpenedImage = InputImage + k \times Mask$ (use $k = 1.5$).
 - Display both the isolated mask and the final sharpened image. Briefly explain why vertical edges are sharpened while horizontal edges remain unaffected.

Task 2: Image Enhancement Transformations (6 Marks)

Select and apply the appropriate spatial enhancement technique for three input images (image_two.jpg, image_three.jpg and image_four.jpg ( Assignment-2 resources)).

2.1 Implementation & Selection (6 Marks):

- Analyse the contrast and intensity distribution of each provided image and apply one of the following techniques per image:
 - Gamma Correction with $\gamma < 1$
 - Gamma Correction with $\gamma > 1$
 - Histogram Equalization

Task 3: Edge-Preserving Filtering for Color Images (8 Marks)

Use image_five.jpg and image_six.jpg ( Assignment-2 resources)

- **3.1 Bilateral Filtering on Colour Image (3 Marks):**
 - Apply bilateral filtering to image_five (noisy RGB colour image). Compute spatial domain weights using spatial distance and range weights.
- **3.2 Joint Bilateral Filtering (3 Marks):**
 - Apply joint bilateral filtering to image_five using image_six as the guidance image. Compute range weights using intensity differences from image_six.
- **3.3 Results and Analysis (2 Marks):**
 - Display the results from both filtering approaches and answer the following and explain the difference in the results of bilateral filtering and joint bilateral filtering.

Link to download the inputs for all tasks:

[Assignment 2 Resources](#)

Submission Guidelines

1. *The content that you submit must be your individual work.*
2. *Submit your code in .py as well as in .ipynb file formats. Both these file submissions are required to receive credit for this assignment*
3. *Ensure your code is well-commented and easy to follow. You can write your answers and explanations using text cells in the Jupyter Notebook files wherever required.*
4. *Submit output files wherever instructed.*
5. *The files should be named as "<roll_number>_assignment_2. For example, if your roll number is 23AI91R01, the code the required file names will be 23AI91R01_assignment_2.py and 23AI91R01_assignment_2.ipynb. You should place all these files within a single zip file and upload it to Moodle as 23AI91R01_assignment_2.zip. Do not use .rar files for submission.*
6. *All submissions must be made through Moodle before the deadline. The submission portal will close at the specified time, and submissions via email will not be accepted.*

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*If you have any queries regarding Assignment 2, please email
Chandra Sekhar Lokesh Manem .*